Representation Theory of Algebras and Quivers

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Abstract

Contents

1 Basic Definitions of Representations

1.1 Representation of an Algebra

Definition 1.1.1 (Representation of an Algebra)
Let R be a commutative ring. Let A be an R-algebra. A representation of A is a left R-module M and a homomorphism of R-algebras

$$\rho : A \to \text{End}_R(M)$$

Let A, M be sets. Recall that there is a natural bijection

$$\text{Hom}_{\textbf{Sets}}(A \times M, M) \cong \text{Hom}_{\textbf{Sets}}(A, \text{Hom}_{\textbf{Sets}}(M, M))$$

given by curling. A representation of an R-algebra A is an element in

$$\text{Hom}_{\textbf{Alg}_R}(A, \text{End}_R(M)) \subseteq \text{Hom}_{\textbf{Sets}}(A, \text{Hom}_{\textbf{Sets}}(M, M))$$

Therefore we may ask what kinds of maps of the form $A \times M \to M$ give a representation of A.

Proposition 1.1.2
Let R be a commutative ring. Let A be an R-algebra. Let M be an R-module. Let $f : A \times M \to M$ be a function. Then the map $a \mapsto f(a, -)$ defines a representation of A if and only if the following are true.

- f is an R-module homomorphism.
- f defines an A-module structure on M.

The A-module structure can also be seen in the representation $\rho : A \to \text{End}_R(M)$. It is given by $a \cdot m = \rho(a)(m)$. In particular, not every A-module M gives a representation; the action of A on M must be bilinear in the scalars R. Therefore these modules are often called R-linear A-modules.

However, if $R = k$ is a field and the structure map $k \to A$ is non-zero, then $k \to A$ is necessarily injective so that A contains k as a subring. Then the k-linearity of $f : A \times M \to M$ is null so that A-modules are the same as representations of A over k.

Corollary 1.1.3
Let k be a field. Let A be a k-algebra. Let V be a k-vector space. Let $f : A \times V \to V$ be a function. Then the map $a \mapsto f(a, -)$ is a representation of A if and only if f defines an A-module structure on V.

Since representations of an algebra A are A-modules, it follows that the following notions in Rings and Modules apply to representations.

- Subrepresentations.
- Quotient Representations.
- Sum of Representations.

Definition 1.1.4 (Types of Representations)
Let R be a commutative ring. Let A be an R-algebra. Let M be a representation of A.

- We say that M is irreducible if it is simple as an A-module.
- We say that M is semisimple if it is semisimple as an A-module.
- We say that M is indecomposable if it is indecomposable as an A-module.

Example 1.1.5
Let k be a field. Define a representation $\rho : k[x] \to \text{End}_k(k^2) \cong M_2(k)$ by $x \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ then extended it R-linearly and respecting products of x. Then ρ is indecomposable but reducible.

Proof
Suppose for a contradiction that ρ is decomposable. Then there exists 1-dimensional linear subspaces L_1, L_2 of k^2 such that ρ_{L_1} and ρ_{L_2} are representations. Suppose that $L_1 = k\langle v_1 \rangle$ and $L_2 = k\langle v_2 \rangle$. Then $x \cdot v_1 \in L_1$ and $x \cdot v_2 \in L_2$ implies that v_1 and v_2 are linearly independent eigenvectors of $\rho(x)$. But by an easy computation, we see that the only eigenvalue of $\rho(x)$ is 1 and its eigenspace has dimension 1. Hence this is a contradiction.

From the above, we see that the eigenspace of $\rho(x)$ is a subrepresentation of ρ. Hence ρ is reducible. ■

Definition 1.1.6 (The Regular Representation)
Let k be a field. Let A be a k-algebra. Define the regular representation $\text{reg} : A \to \text{End}_k(A)$ of A by
$$\text{reg}(a)(b) = ab$$

Definition 1.1.7 (Composition Factors)
Let R be a commutative ring. Let A be an R-algebra. Let M be a semisimple representation of A. Suppose that the irreducible decomposition of M is given by
$$M \cong \bigoplus_{i=1}^{n} M_i$$
Define the composition factors of M to be
$$\text{Comp}(M) = \{M_1, \dots, M_n\}$$

Definition 1.1.8 (Multiplicity of an Irreducible Submodule)
Let R be a commutative ring. Let A be an R-algebra. Let M be a semisimple representation of A. Let N be an irreducible A-module. Define the multiplicity of N in M to be
$$\text{mult}_N(M) = |\{M_i \in \text{Comp}(M) \mid M_i \cong N\}|$$

1.2 Maps between Representations

Definition 1.2.1 (Homomorphisms of Representations)
Let R be a commutative ring. Let A be an R-algebra. Let M, N be representations of A. Let $f : M \to N$ be a function. We say that f is a homomorphism of representations if the following are true.
- f is an A-module homomorphism.
- f is R-linear.

Since M and N are A-modules, the definition of homomorphism implies that such a morphism must be an A-module homomorphism. Conversely, since not every A-module is an R-linear A-module, A-module homomorphisms are not necessarily morphisms between representations.

Definition 1.2.2 (Set of Homomorphisms of Representations)
Let R be a commutative ring. Let A be an R-algebra. Let M, N be representations of A. Define the set of homomorphisms from M to N to be
$$\text{Hom}_{A_R}(M, N) = \left\{ f : M \to N \;\middle|\; \substack{f \text{ is a homomorphism} \\ \text{of representations from } M \text{ to } N} \right\} \subseteq \text{Hom}_A(M, N)$$

Definition 1.2.3 (Isomorphisms of Representations) Let R be a commutative ring. Let A be an R-algebra. Let M,N be R-modules. Let $\rho : A \to \text{End}_R(M)$ and $\tau : A \to \text{End}_R(N)$ be representations of A. Let $\phi : M \to N$ be a homomorphism of representations. We say that ϕ is an isomorphism if there exists a homomorphism of representations $\psi : N \to M$ such that

$$\psi \circ \phi = \text{id}_M \quad \text{and} \quad \phi \circ \psi = \text{id}_N$$

Lemma 1.2.4
Let R be a commutative ring. Let A be an R-algebra. Let M,N be R-modules. Let $\rho : A \to \text{End}_R(M)$ and $\tau : A \to \text{End}_R(N)$ be representations of A. Let $\phi : M \to N$ be a homomorphism of representations. Then ϕ is an isomorphism if and only if ϕ is an isomorphism of R-modules.

Theorem 1.2.5 (First Isomorphism Theorem for Representations)
Let R be a commutative ring. Let A be an R-algebra. Let M,N be R-modules. Let $\rho : A \to \text{End}_R(M)$ and $\tau : A \to \text{End}_R(N)$ be representations of A. Let $\phi : M \to N$ be a homomorphism of representations. Then the following are true.

- $\ker(\phi)$ is a subrepresentation of M.
- $\text{im}(\phi)$ is a subrepresentation of N.
- There is an isomorphism of representations

  $$M/\ker(\phi) \cong \text{im}(\phi)$$

  given by ϕ.

Proposition 1.2.6
Let R be a commutative ring. Let A be a finite R-algebra. Let $\rho : A \to \text{End}_R(M)$ be an irreducible representation of A. Then M is finitely generated.

Proof
Let $0 \neq m \in M$. Then $\rho(A)(m)$ is a submodule of M. Since M is irreducible, we must have $\rho(A)(m) = M$. Since A is finite over R, we must have $\rho(A)(m)$ is finitely generated over R. Hence M is finitely generated over R. ∎

1.3 The Set of Representations

Definition 1.3.1 (The Set of Representations)
Let R be a commutative ring. Let A be an R-algebra. Let M be an R-module. Define the space of representations of A on M to be the set

$$\text{Rep}(A,M) = \{[\rho : A \to \text{End}_R(M)] \mid [\rho] \text{ is an isomorphism class of representations }\}$$

2 Finite Dimensional Representations of Algebra over Algebraically Closed Fields

2.1 Schur's Lemma and its Consequences

Recall from Rings and Modules that we have Schur's lemma: A homomorphism $M \to N$ of simple R-modules is either the 0 map or an isomorphism.

Lemma 2.1.1 (Schur's Lemma III)
Let k be an algebraically closed field. Let A be a k-algebra. Let V be a finite dimensional k-vector space. Let $\rho : A \to \text{End}_k(V)$ be an irreducible representations of A. Let $\phi : V \to V$ be a homomorphism of representations. Then $\phi = \lambda\text{id}_V$ for some $\lambda \in k$.

Proof
Since ϕ is a homomorphism of simple A-modules, by Schur's lemma we know that $\phi = 0$ or ϕ is an isomorphism. If $\phi = 0$ then we are done. Otherwise, since k is algebraically closed, the endomorphism ϕ admits an eigenvalue $\lambda \in k$ and eigenvector $v \in V$. Consider the homomorphism of representations $\phi - \lambda\text{id}_V$. By Schur's lemma, this map is either 0 or an isomorphism. But it is not an isomorphism since $\phi(v) - \lambda v = 0$. Hence $\phi - \lambda\text{id}_V = 0$. So we are done. ■

Corollary 2.1.2
Let k be an algebraically closed field. Let A be a commutative k-algebra. Let V be a finite dimensional k-vector space. Let $\rho : A \to \text{End}_k(V)$ be an irreducible representations of A. Then $\dim(V) = 1$.

Proof
By Schur's lemma III, we know that $\rho(a) : V \to V$ is a scalar operator for all $a \in A$. This means that $\rho(a)(v) = \lambda_a v$ for some $\lambda_a \in k$ and all $v \in V$. For any fixed $v_0 \in V$, this implies that $k\langle v_0\rangle$ is a subrepresentation of ρ. If $\dim(V) \geq 2$, then $k\langle v_0\rangle$ is a non-trivial subrepresentation, contradicting the irreducible assumption. Hence $\dim(V) = 1$. ■

In other words, every finite dimensional representation of a commutative algebra over an algebraically closed field must be of dimension 1. And these are easy to classify.

Proposition 2.1.3
Let k be an algebraically closed field. Let A be a k-algebra. Let V, W be finite dimensional irreducible representations of A. Then we have

$$\dim_k(\text{Hom}_{A_k}(V, W)) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Proof
If V is not isomorphic to W, then by Schur's lemma we must have $\text{Hom}_{\mathbf{Rep}(A)}(V, W) \subseteq \text{Hom}_A(V, W) = \{0\}$. If $V = W$, then Schur's lemma III implies that id_V generates the k-vector space. ■

Proposition 2.1.4
Let k be an algebraically closed field. Let A be a k-algebra. Let V_i, W_j be finite dimensional irreducible representations of A for $1 \leq i \leq n$ and $1 \leq j \leq m$. Then we have

$$\dim_k\left(\text{Hom}_{A_k}\left(\bigoplus_{i=1}^n V_i, \bigoplus_{j=1}^m W_j\right)\right) = \sum_{i=1}^n \sum_{j=1}^m \dim_k(\text{Hom}_{A_k}(V_i, W_j))$$

Corollary 2.1.5
Let k be an algebraically closed field. Let A be a k-algebra. Let V, W be finite dimensional representations of A such that W is irreducible. Then we have

$$\dim_k(\text{Hom}_{A_k}(V, W)) = \text{mult}_W(V)$$

2.2 Character Theory

Let k be an algebraically closed field. Recall that the trace of a k-linear map between finite dimensional vector spaces have a basis independent description.

Definition 2.2.1 (Characters of a Representation)
Let k be an algebraically closed field. Let A be a k-algebra. Let $\rho : A \to \text{End}_k(V)$ be a finite dimensional representation of A. Define the character of ρ to be function $\chi_V : A \to k$ given by

$$\chi_V(a) = \text{tr}(\rho(a))$$

Proposition 2.2.2
Let k be an algebraically closed field. Let A be a k-algebra. Let V, W be isomorphic finite dimensional representations of A. Then $\chi_V = \chi_W$.

Proposition 2.2.3
Let k be an an algebraically closed field. Let A be a k-algebra. Let $\rho : A \to \text{End}_k(V)$ be a finite dimensional representation of A. Then χ_V descends to a well defined function $\chi_V : \frac{A}{[A,A]} \to k$.

Let G be a finite group. Since $\mathbb{C}[G]$ is a semisimple $\mathbb{C}$-algebra, we know that there are $\mathbb{C}$-vector space isomorphisms

$$\begin{aligned}
\text{Hom}_{\mathbb{C}}\left(\frac{\mathbb{C}[G]}{[\mathbb{C}[G], \mathbb{C}[G]]}, \mathbb{C}\right) &= \{f \in \text{Hom}_{\mathbb{C}}(\mathbb{C}[G], \mathbb{C}) \mid f(ghg^{-1}h^{-1}) = 0 \text{ for all } g, h \in G\} \\
&= \{f \in \text{Hom}_{\textbf{Set}}(G, \mathbb{C}) \mid f(ghg^{-1}) = f(h) \text{ for all } g, h \in G\} \\
&= \text{Hom}_{\textbf{Set}}(\text{Cl}(G), \mathbb{C})
\end{aligned}$$

Hence the above proposition mimics character theory in representations of finite groups.

Lemma 2.2.4
Let k be an algebraically closed field. Let A be a k-algebra. Let V be a finite dimensional representation of A. Let $W \leq V$ be a subrepresentation. Then we have

$$\chi_V = \chi_W + \chi_{V/W}$$

Definition 2.2.5 (Irreducible Characters)
Let k be an algebraically closed field. Let A be a k-algebra. Let V be a representation of A. We say that χ_V is irreducible if V is irreducible.

Proposition 2.2.6
Let k be an algebraically closed field. Let A be a k-algebra. Let $V_1, \dots, V_n$ be finite dimensional irreducible representations of A. Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ are linearly independent in $\text{Hom}_k(A/[A, A], k)$.

Definition 2.2.7 (Characters of an Algebra)
Let k be an algebraically closed field. Let A be a k-algebra. Let $\phi : A \to k$ be a function. We say that ϕ is a character of A if there exists a finite dimensional representation V of A such that $\phi = \chi_V$.

2.3 The Semisimple Condition

Proposition 2.3.1
Let k be an algebraically closed field. Let A be a finite semisimple k-algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic representations of A. Then there is a k-algebra isomorphism

$$A \cong \bigoplus_{i=1}^{n} \text{End}_k(V_i)$$

Proof
By the Artin Wedderburn theorem, there is a k-algebra isomorphism

$$A \cong \bigoplus_{i=1}^{p} M_{n_i}(D_i)$$

where each D_i is a division algebra over k. Since k is algerbaically closed, we have $D_i = k$. By a corollary of the Artin-Wedderburn theorem, the complete list of simple A-modules is given by the unique simple $M_{n_i}(k)$-submodule T_i of itself. Hence $n = p$. Moreover, by reordering the index, we have isomorphisms of $M_{n_i}(k)$-modules $V_i \cong T_i$ that are isomorphisms of A-modules (since the ring structure of the product of matrices is given by the product of the ring structures of the matrix ring, and the Artin-Wedderburn isomorphism is a ring isomorphism). Now $T_i \cong k^{n_i}$ as k-vector spaces. This gives an isomorphism $V_i \cong k^{n_i}$ of k-vector spaces and hence an isomorphism of rings $\text{End}_k(V_i) \cong M_{n_i}(k)$. Thus we have

$$A \cong \bigoplus_{i=1}^{p} M_{n_i}(k) \cong \bigoplus_{i=1}^{n} \text{End}_k(V_i)$$

■

Corollary 2.3.2
Let k be an algebraically closed field. Let A be a finite semisimple k-algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic representations of A. Then we have

$$\dim_k(A) = \sum_{i=1}^{n} \dim_k(V_i)^2$$

Corollary 2.3.3
Let k be an algebraically closed field. Let A be a finite k-algebra. Let V be an irreducible representation of A. The we have

$$\text{mult}_V(A) = \dim_k(\text{Hom}_{A_k}(A, V)) = \dim_k(V)$$

Proof
Direct from Artin-Wedderburn theorem and its corollaries. ■

Proposition 2.3.4
Let k be an algebraically closed field. Let A be a finite semisimple k-algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible finite dimensional representation of A. Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ is a basis for $\text{Hom}_k(A/[A, A], k)$.

Corollary 2.3.5
Let k be an algebraically closed field. Let A be a finite semisimple k-algebra. Let χ be a character of A. Then there exists a representation V of A such that $\chi = \chi_V$.

2.4 Reduction to Indecomposable Basic Algebras

Let A be a finite dimensional k-algebra over a field k. Recall that $A^\flat = eAe$ is the basic algebra associated to A, where $e = \sum_{i=1}^n e_i$ and $e_1, \dots, e_n$ is a full orthogonal system of primitive idempotents.

Proposition 2.4.1
Let k be an algebraically closed field. Let A be a finite dimensional k-algebra. Then there is an equivalence of categories

$$_A\textbf{FGMod} \simeq {}_{A^\flat}\textbf{FGMod}$$

given by the restriction and extension of scalars.

Proposition 2.4.2
Let k be an algebraically closed field. Let A, B be k-algebras. Let V be a finite dimensional irreducible representation of $A \times B$. Then there exists irreducible representations V_A and V_B of A and B respectively, such that $V \cong V_A \otimes_k V_B$.

3 Quivers and Path Algebras

3.1 Basics of Quivers

Definition 3.1.1 (Quivers)
A quiver consists of two sets V and E called vertices and edges, together with two functions $s, t : E \to V$ called the source and target respectively.

Definition 3.1.2 (Finite Quivers)
Let $Q = (V, E, s, t)$ be a quiver. We say that Q is finite if V and E are finite.

Definition 3.1.3 (Connected Quivers)

Definition 3.1.4 (Paths and Stationary Path)
Let $Q = (V, E, s, t)$ be a quiver. Let $a, b \in V$. A path from a to b of length $n \in \mathbb{N}^+$ is a function $\gamma : \{1, \dots, n\} \to E$ such that the following are true.

- $s(\gamma(1)) = a$.
- $s(\gamma(i + 1)) = t(\gamma(i))$ for $1 \leq i \leq n - 1$.
- $t(\gamma(n)) = b$.

For each $v \in V$, we associate a path ε_v of length 0 called the stationary path at v.

Definition 3.1.5 (Cycles)
Let $Q = (V, E, s, t)$ be a quiver. Let $\gamma : \{1, \dots, n\} \to E$ be a path in Q. We say that γ is a cycle if $s(\gamma(1)) = t(\gamma(n))$.

Definition 3.1.6 (Acyclic Quivers)
Let Q be quiver. We say that Q is acyclic if Q contains no cycles.

3.2 Path Algebra

Definition 3.2.1 (The Path Algebra of a Quiver)
Let Q be a quiver. Let k be a field. Define the path algebra $k[Q]$ of Q over k to be the k-vector space

$$k\langle Q\rangle = k\langle \gamma \mid \gamma \text{ is a path in } Q\rangle$$

together with multiplication given as follows. For two paths $\gamma_1 : \{1, \dots, n_1\} \to E$ and $\gamma_2 : \{1, \dots, n_2\} \to E$, define $\gamma_1 \cdot \gamma_2 : \{1, \dots, n_1 + n_2\} \to E$ by

$$(\gamma_1 \cdot \gamma_2)(i) = \begin{cases} \gamma_1(i) & \text{if } 1 \leq i \leq n_1 \\ \gamma_2(i - n_1) & \text{if } n_1 + 1 \leq i \leq n_2 \end{cases}$$

if $t(\gamma_1(n_1)) = s(\gamma_2(1))$ and 0 otherwise.

Lemma 3.2.2
Let $Q = (V, E, s, t)$ be a quiver. Let k be a field. If V is finite, then the following are true.

- $k\langle Q\rangle$ is a graded k-algebra whose grading is given by the length of the paths.
- We have $1_{k\langle Q\rangle} = \sum_{v \in V} \varepsilon_v$.
- $\{\gamma : \{1\} \to E \mid \gamma \text{ is a path in } Q\}$ is a generating set for $k\langle Q\rangle$ as a module over itself.
- $\{\varepsilon_v \mid v \in V\}$ forms a full orthogonal system of primitive idempotents for $k\langle Q\rangle$.

Example 3.2.3
Let $Q = (V, E, s, t)$ be the quiver given by $V = \{v\}$ and $E = \{\alpha_i \mid i \in I\}$. Then we have a k-algebra

isomorphism
$$k\langle Q\rangle \cong k\langle \alpha_i \mid i \in I\rangle$$
to the free associative algebra generated by E.

Proposition 3.2.4
Let Q be a quiver. Let k be a field. Then the following are true.
- $k\langle Q\rangle$ is finite dimensional over k if and only if Q is finite and acyclic.
- $k\langle Q\rangle$ is indecomposable if and only if Q is connected.

Proposition 3.2.5 (Universal Property of Finite Connected Quivers Algebras)
Let $Q = (V, E, s, t)$ be a finite connected quiver. Let k be a field. Then $k\langle Q\rangle$ and the inclusion functions $\iota_V : V \to k\langle Q\rangle$ and $\iota_E : E \to k\langle Q\rangle$ satisfies the following universal property: For any k-algebra A and any maps $\varphi_V : V \to A$ and $\varphi_E : E \to A$ such that the following are true:
- $1 = \sum_{v\in V} \varphi_V(v)$, $\varphi_V(v)^2 = \varphi_V(v)$ and $\varphi_V(v)\varphi_V(w) = 0$ for all $v, w \in V$ and $v \neq w$.
- We have $\varphi_E(p) = \varphi_V(s(p))\varphi_E(p)\varphi_V(t(p))$ for all $p \in E$.

there exists a unique k-algebra homomorphism $\varphi : k\langle Q\rangle \to A$ such that the following diagram commutes:
$$\begin{tikzcd}
V \arrow[r, "\iota_V"] \arrow[dr, "\varphi_V"'] & k\langle Q\rangle \arrow[d, dashed, "\exists!\varphi"'] & E \arrow[l, "\iota_E"'] \arrow[dl, "\varphi_E"] \\
& A &
\end{tikzcd}$$
Moreover, $k\langle Q\rangle$ is the unique (up to unique isomorphism) k-algebra with the above property.

Definition 3.2.6 (The Arrow Ideal)
Let Q be a finite quiver. Let k be a field. Define the arrow ideal of the path algebra $k\langle Q\rangle$ to be
$$R_Q = k\langle \gamma \mid \gamma \text{ is a path in } Q \text{ of length at least } 1\rangle$$

Lemma 3.2.7
Let Q be a finite quiver. Let k be a field. Then the following are true.
- R_Q is a two sided graded ideal of $k\langle Q\rangle$.
- The quotient of the path algebra by the arrow ideal is given by
$$k\langle Q\rangle/R_Q = k\langle \varepsilon_v \mid v \in V\rangle$$
- For each $n \in \mathbb{N}^+$, the product of the arrow ideal is given by
$$R_Q^n = k\langle \gamma \mid \gamma \text{ is a path in } Q \text{ of length at least } n\rangle$$
- For each $n \in \mathbb{N}^+$, we have
$$R_Q^n/R_Q^{n+1} = k\langle \gamma \mid \gamma \text{ is a path in } Q \text{ of length } n\rangle$$

Proposition 3.2.8
Let Q be a finite quiver. Let k be a field. If Q is acyclic, then the following are true.
- $J(k\langle Q\rangle) = R_Q$.
- $k\langle Q\rangle$ is a finite dimensional basic algebra over k.

Definition 3.2.9 (Admissible Ideals)
Let Q be a finite quiver. Let k be a field. Let I be an ideal of $k\langle Q\rangle$. We say that I is admissible if there exists $n \geq 2$ such that
$$R_Q^n \subseteq I \subseteq R_Q^2$$

Definition 3.2.10 (Bound Quiver Algebra)
Let Q be a finite quiver. Let k be a field. Let I be an ideal of $k\langle Q\rangle$. We say that $k\langle Q\rangle/I$ is a bound quiver algebra if I is an admissible ideal.

Proposition 3.2.11
Let $Q=(V,E,s,t)$ be a finite quiver. Let k be a field. Let I be an admissible ideal of $k\langle Q\rangle$. Then $\{\varepsilon_v+I \mid v\in V\}$ is a full orthogonal system of primitive idempotents for the bound quiver algebra $k\langle Q\rangle/I$.

Proposition 3.2.12
Let Q be a finite quiver. Let k be a field. Let I be an admissible ideal of $k\langle Q\rangle$. Then the following are true.

- I is finitely generated.
- $J(k\langle Q\rangle/I)=R_Q/I$.
- $k\langle Q\rangle/I$ is a finite dimensional basic algebra over k.
- $k\langle Q\rangle/I$ is indecomposable if and only if $k\langle Q\rangle$ is indecomposable.

3.3 The Quiver of a Finite Dimensional Basic Algebra

Definition 3.3.1 (The Quiver of a Basic Algebra)
Let k be a field. Let A be a finite dimensional basic algebra over k. Let $e_1,\dots,e_n$ be a full orthogonal system of primitive idempotents. Define the quiver $Q_A=(V,E,s,t)$ associated to A as follows.

- The vertices are given by $V=\{e_1,\dots,e_n\}$.
- For $e_i,e_j\in V$, the set of arrows from e_i to e_j are given by $e_i\frac{J(A)}{J^2(A)}e_j$.

Lemma 3.3.2
Let k be a field. Let A be a finite dimensional basic algebra over k. Then Q_A does not depend on the choice of full orthogonal system of primitive idempotents.

Theorem 3.3.3 (Gabriel)
Let k be an algebraically closed field. Let A be a finite dimensional basic algebra over k. Then there exists an admissible ideal I of $k\langle Q_A\rangle$ such that there is a k-algebra isomorphism

$$A\cong\frac{k\langle Q_A\rangle}{I}$$

Corollary 3.3.4
Let k be an algebraically closed field. Let A be a finite dimensional algebra over k. Then A is Morita equivalent to $k\langle Q_{A^\flat}\rangle/I$ where I is an admissible ideal of $k\langle Q_{A^\flat}\rangle$.

Thus the representation theory of a finite dimensional algebra over an algebraically closed field is equivalent to the representation theory of bound finite quiver algebras.

4 Representation Theory of Quivers

4.1 Representations of Quivers

Definition 4.1.1 (Representations of Quivers)
Let $Q = (V, E, s, t)$ be a quiver. Let R be a ring. A representation of Q over R consists of the following.

- A collection of R-modules M_v for each $v \in V$.
- A collection of R-module homomorphisms $\varphi_e : M_{s(e)} \to M_{t(e)}$ for each $e \in E$.

Note that the resulting diagram of R-modules need not be commutative because we have not imposed any conditions on commutativity.

Definition 4.1.2 (Homomorphism of Representations)

Definition 4.1.3 (The Category of Representations of a Quiver)

Definition 4.1.4 (The Category of Representations of a Quiver over an Admissible Ideal)
Let k be a field. Let Q be a quiver. Let I be an admissible ideal of the path algebra $k\langle Q\rangle$. Define $\mathbf{Rep}(Q, I)$ to be the full subcategory of $\mathbf{Rep}(Q)$ consisting of objects $((V_i)_{i\in V}, (T_j)_{j\in E})$ such that $V_k = 0$ for all $k \in I$.

4.2 Relation to Representations of the Path Algebra

Proposition 4.2.1
Let Q be a quiver. Let k be a field. Then there is a natural isomorphism

$$\mathbf{Rep}(Q) \cong {}_{k\langle Q\rangle}\mathbf{Mod}$$

Proposition 4.2.2
Let Q be a finite quiver. Let k be a field. Let I be an admissible ideal of $k\langle Q\rangle$. Then there is an equivalence of categories

$$\mathbf{Rep}(Q, I) \simeq {}_{k\langle Q\rangle/I}\mathbf{FGMod}$$

Corollary 4.2.3
Let k be an algebraically closed field. Let A be a finite dimensional k-algebra. Then there exists an admissible ideal I of $Q_{A^\flat}$ such that there is an equivalence of categories

$${}_A\mathbf{FGMod} \simeq \mathbf{Rep}(Q_{A^\flat}, I)$$